

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

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Experiment No.: 5-1

1. TITLE

Interactive Product Calculator

2. AIM

To create an interactive webpage containing a product name, quantity, and price, and dynamically calculate the total price whenever the quantity or price changes without refreshing the page.

3. OBJECTIVE

The objectives of this experiment are:

- To understand JavaScript functions and variables.
 - To use DOM manipulation to access and update HTML elements.
 - To use event listeners to respond to changes in user input.
 - To calculate the total price dynamically.
 - To validate quantity and price values.
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4. THEORY

JavaScript is a scripting language used to make web pages interactive by responding to user actions and changing webpage content dynamically.

The Document Object Model (DOM) represents an HTML document as objects that JavaScript can access and modify. In this application, JavaScript obtains the quantity and price values from the input elements, multiplies them, and displays the result in the total-price element.

Event listeners detect changes in the quantity and price fields, so the calculation is updated immediately without reloading the webpage.

Formula: Total Price = Quantity × Price per Unit

5. ALGORITHM / PROCEDURE

1. Create an HTML webpage with fields for product name, quantity, and price.
 2. Create an element to display the total price.
 3. Define a JavaScript function named `calculateTotal()`.
 4. Read the quantity and price values from the DOM.
 5. Validate the quantity and price values.
 6. Multiply quantity by price.
 7. Display the calculated total in the webpage.
 8. Attach input event listeners to the quantity and price fields.
 9. Call `calculateTotal()` when the page loads.
 10. Open the webpage in a browser and test different inputs.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Interactive Product Calculator</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="calculator">
    <h1>Product Calculator</h1>

    <div class="form-group">
      <label for="productName">Product Name:</label>
      <input type="text" id="productName" placeholder="Enter product name">
    </div>

    <div class="form-group">
      <label for="quantity">Quantity:</label>
      <input type="number" id="quantity" value="1" step="any">
    </div>

    <div class="form-group">
      <label for="price">Price per Unit (₹):</label>
      <input type="number" id="price" value="0" step="any">
    </div>

    <p id="errorMessage" class="error"></p>

    <div class="result">
      <strong>Total Price:</strong>
      <span id="totalPrice">₹0.00</span>
    </div>
  </div>

  <script src="script.js"></script>
</body>
</html>
```

style.css

```
* {
  box-sizing: border-box;
}

body {
  margin: 0;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  font-family: Arial, "DejaVu Sans", sans-serif;
  background-color: #f2f2f2;
}

.calculator {
  width: 400px;
  padding: 30px;
  background-color: white;
  border: 1px solid #ccc;
  border-radius: 8px;
}

h1 {
  margin-top: 0;
  text-align: center;
}
```

```

.form-group {
  margin-bottom: 18px;
}

label {
  display: block;
  margin-bottom: 6px;
  font-weight: bold;
}

input {
  width: 100%;
  padding: 10px;
  font-size: 16px;
  border: 1px solid #aaa;
  border-radius: 4px;
}

input:focus {
  outline: none;
  border-color: #000;
}

.error {
  min-height: 40px;
  margin: 0 0 10px;
  color: #b00020;
  font-size: 14px;
}

.result {
  display: flex;
  justify-content: space-between;
  padding: 12px;
  border: 1px solid #ccc;
  border-radius: 4px;
  font-size: 18px;
}

```

6. PROGRAM (CONTINUED)

script.js

```

const quantityInput = document.getElementById("quantity");
const priceInput = document.getElementById("price");
const totalDisplay = document.getElementById("totalPrice");
const errorDisplay = document.getElementById("errorMessage");

function calculateTotal() {
  const quantity = parseFloat(quantityInput.value);
  const price = parseFloat(priceInput.value);

  errorDisplay.textContent = "";

  if (isNaN(quantity) || isNaN(price) || quantity < 0 || price < 0) {
    errorDisplay.textContent =
      "Please enter valid non-negative values for quantity and price.";
    totalDisplay.textContent = "₹0.00";
    return;
  }

  const total = quantity * price;
  totalDisplay.textContent = "₹" + total.toFixed(2);
}

quantityInput.addEventListener("input", calculateTotal);
priceInput.addEventListener("input", calculateTotal);

calculateTotal();

```

7. CODE EXPLANATION

The program is a single web page made of three files. **index.html** creates the form with the product name, quantity and price fields, an error-message paragraph and the total-price display. **style.css** centres the calculator card and styles the fields. **script.js** holds the logic: the four constants at the top store references to the page elements, and the function **calculateTotal()** reads the two numeric inputs, checks them, multiplies them and writes the formatted result to the page. Because the function is attached to the input event of both fields, the total is recomputed on every keystroke, and it is also called once when the page loads so that the initial total is shown.

Code	Explanation
const quantityInput, priceInput, totalDisplay, errorDisplay	Store references to the four HTML elements once, so the function can reuse them.
function calculateTotal()	Defines the function used to calculate and display the total price.
getElementById()	Accesses HTML elements using their IDs.
parseFloat()	Converts the text typed in an input field into a decimal number (NaN if the field is empty).
if (isNaN(...) ... < 0)	Checks whether quantity and price are valid non-negative numbers; if not, shows an error message and sets the total to ₹0.00.
quantity * price	Calculates the total price.
toFixed(2)	Formats the total with exactly two decimal places (for example 89.25).
textContent	Updates the total price and the error message shown on the webpage.
addEventListener("input", ...)	Runs the calculation whenever the quantity or price changes.
calculateTotal();	Performs the initial calculation when the page loads.

8. TEST CASES AND EXECUTION DATA

Test Case	Product	Quantity	Price	Expected Result	Actual Result	Status
TC-01	Marker	12	₹35	Total = ₹420.00	Total = ₹420.00	Pass
TC-02	Stapler	3	₹85.50	Total = ₹256.50	Total = ₹256.50	Pass
TC-03	Eraser	0	₹8	Total = ₹0.00	Total = ₹0.00	Pass
TC-04	Ruler	7	₹12.75	Total = ₹89.25	Total = ₹89.25	Pass

Additional Validation Test

Test Case	Input	Expected Result	Actual Result	Status
TC-05	Negative quantity (Compass, quantity -4, price ₹30)	Validation message should be displayed.	Validation message displayed; total = ₹0.00	Pass
TC-06	Negative price (Sharpener, quantity 6, price -₹15)	Validation message should be displayed.	Validation message displayed; total = ₹0.00	Pass

9. OUTPUT

Output 1: Product Calculator with Entered Values (TC-01: Marker, quantity 12, price ₹35)

Product Calculator

Product Name:

Quantity:

Price per Unit (₹):

Total Price: ₹420.00

Output 2: Product Calculator Initial State

Product Calculator

Product Name:

Quantity:

Price per Unit (₹):

Total Price: ₹0.00

Output 3: Validation Message for Negative Quantity (TC-05: Compass, quantity -4, price ₹30)

Product Calculator

Product Name:

Quantity:

Price per Unit (₹):

Please enter valid non-negative values for quantity and price.

Total Price: ₹0.00

10. RESULT

Thus, the Interactive Product Calculator was implemented using HTML, CSS, and JavaScript. The output screenshots verify that a quantity of 12 and a unit price of ₹35 produce a total price of ₹420.00 without refreshing the webpage, and that negative values display a validation message.

How this would be submitted

Source Code

```
Q1_Interactive_Product_Calculator/  
├─ index.html  
├─ style.css  
├─ script.js  
└─ README.md
```